ModelIt!

Teacher's Guide

Where Does a Tree's Mass Come From?

A 1-Ton Tree Got Most of Its Weight From Air

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-LS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-LS1-1

Support an argument that plants get the materials they need for growth chiefly from air and water.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students construct an explanation for where plant mass comes from.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow** — Plants acquire material for growth chiefly from air and water.
Crosscutting Concept	**Matter and Energy** — Students trace matter from air into plant biomass.

CAST Testing Alignment

MULTIPLE CHOICE:

Where do trees get most of the mass they need to grow?

CONSTRUCTED RESPONSE:

A greenhouse farmer wants maximum plant growth. Based on your model, what two things should she increase?

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The discovery that plants get mass from air, not soil, is one of the most counterintuitive findings in biology. Van Helmont's 1648 experiment began the investigation.

Van Helmont's Experiment

In 1648, Van Helmont grew a willow tree for 5 years in a weighed pot of soil. The tree gained 164 pounds while the soil lost only 2 ounces. He concluded (partially correctly) that the mass came from water.

Photosynthesis Discovery

Later scientists discovered the complete picture: $6CO_2 + 6H_2O$ + light energy --> $C_6H_{12}O_6 + 6O_2$. Carbon dioxide from air provides the carbon atoms that become plant tissue.

Carbon Storage

Trees are essentially solidified carbon dioxide. When we cut down forests, that stored carbon returns to the atmosphere as CO_2.

Climate Connection

The Amazon rainforest stores about 90 billion tons of carbon. Deforestation releases this stored carbon, accelerating climate change.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify CO_2, sunlight, photosynthesis rate, and tree biomass as system components.

E — Establish Relationships
Students establish that both CO_2 and sunlight positively affect photosynthesis, which positively affects tree mass.

V — Visualize & Model
Students build the tree growth model showing how air becomes wood.

E — Evaluate Outcomes
Students run no-CO_2 scenarios and observe tree growth completely stopping.

R — Revise & Extend
Students add water to complete the photosynthesis equation.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This tree weighs 1 million pounds. Where did it come from?"*

Watch For

Take predictions - most will say soil.

Slide 2: Learning Objectives

2 min

Say: *"Today we discover plants' surprising source of mass."*

Watch For

Define key vocabulary.

Slide 3: Big Question

2 min

Say: *"Van Helmont proved it wasn't soil. So what is it?"*

Watch For

Share the 164 lbs vs 2 oz result.

Slide 4: LEVER Framework

2 min

Say: *"We'll model how trees grow."*

Watch For

Preview activities.

Slide 5: Activity 1: Components

8 min

Say: *"CO2 is the raw material - the carbon atoms."*

Watch For

Connect to the air around them.

Slide 6: Activity 2: Connections

8 min

Say: *"Both inputs drive photosynthesis which builds mass."*

Watch For

Discuss why all positive.

Slide 7: Activity 3: Simulation

10 min

Say: *"Remove the CO2. Watch the tree stop growing."*

Watch For

Students run the revealing scenario.

Slide 8: Discoveries

5 min

Say: *"Every tree is made of air that became solid."*

Watch For

Connect to climate change.

Slide 9: STEM Challenge

10 min

Say: *"Design a forest to capture maximum carbon."*

Watch For

Research tree species and CO2 absorption.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** CO2 in Air, Sunlight
- **INTERNAL:** Photosynthesis Rate, Tree Biomass

Relationships

- **CO2 in Air to Photosynthesis Rate:** POSITIVE (+) — More CO2 means more raw material for building sugar.
- **Sunlight to Photosynthesis Rate:** POSITIVE (+) — More light energy powers faster photosynthesis.
- **Photosynthesis Rate to Tree Biomass:** POSITIVE (+) — More photosynthesis produces more glucose which becomes more wood.

Key Discoveries

- Trees get most of their mass from CO2 in the air - not from soil
- Soil provides minerals but not the bulk mass of the tree
- Remove CO2 and tree growth stops completely
- This is why deforestation releases stored carbon back to atmosphere

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants get mass from soil"	Soil provides minerals and water, but the MASS comes from CO_2 in the air. Van Helmont proved this in 1648.
"Trees take in oxygen"	Trees produce oxygen through photosynthesis. They take in CO_2.
"Plants don't need sunlight, just water"	Sunlight is the energy source that powers photosynthesis. No light = no photosynthesis = no growth.